HUGO MOSSER

github.com/strandedDepths | strandedDepths.github.io | linkedin.com/in/hugo-mosser/ | hmosser [dot] p [at] gmail [dot] com |
Pau, France Open to relocation worldwide

SUMMARY

AI/ML Engineer with a French Engineering degree (Master of Engineering) in Computer Science with AI specialization and a track record of shipping AI systems end-to-end: from research prototypes to deployed APIs. Skilled in Python, PyTorch, TensorFlow, FastAPI and cloud-based model serving (GCP, TF Serving), with experience spanning NLP, Computer Vision and LLM integration (LangChain, RAG, OpenAI API). Built and published multiple production-oriented projects independently. International research background (Osaka Metropolitan University, Japan). Open to relocation worldwide.

WORK EXPERIENCE

Volunteer & Founding Full-Stack Engineer — Jun 2026 – Present

FlowSignal · England (Remote)

- Sole engineer responsible for the design, architecture and implementation of the FlowSignal platform using Python/FastAPI, PostgreSQL and React, working closely with the founders on product direction and technical decisions.
- Identified and resolved a critical trust vulnerability through a self-directed security review, strengthening runtime authority validation and supporting the implementation with 52 automated unit, boundary and integration tests.
- Authored key technical documentation, including engineering white papers, Authority Control for Autonomous AI Agents paper, formal specification reviews and implementation documentation supporting the FlowSignal Runtime Authority architecture.

AI Engineer – Independent Projects — Sep 2025 – Present

Self-directed · Remote

- Architected and deployed end-to-end machine learning systems, including a GCP-hosted plant disease classification mobile app (TensorFlow/Keras CNN, FastAPI, React Native) achieving 96% test accuracy, and a hybrid log classification microservice combining Regex, Sentence Transformers, and LLM fallback.
- Developed production-ready AI applications such as a LangChain and OpenAI-powered RAG news research tool with Streamlit. All projects feature fully documented public GitHub repositories and deployable backends, demonstrating full-lifecycle expertise from data modeling to cloud deployment.

AI R&D Engineer Intern — Apr 2024 – Oct 2024

STMicroelectronics · Crolles · France

- Developed neural surrogate models in PyTorch to approximate computationally expensive analog circuit simulations, enabling faster design iteration cycles
- Built end-to-end data preprocessing and training pipelines in Python, handling domain specific engineering data from raw inputs to model-ready tensors
- Evaluated model performance against real-world design cases in collaboration with an engineering team, iterating on architecture and training strategy to meet accuracy requirements

ML/AI Research Scientist Intern — Apr 2023 – Sep 2023

Osaka Metropolitan University · Osaka, Japan

- Conducted research on multimodal AI for onomatopoeia generation: designed a pipeline mapping visual inputs (images) to textual sound representations using deep learning models in TensorFlow
- Implemented and benchmarked image captioning and sequence generation architectures, adapting them to a low-resource cross-lingual setting (Japanese/English)
- Conducted full experimental lifecycle: dataset curation, model training, evaluation, and results analysis in an international academic research environment

TECHNICAL SKILLS

Languages	Python, C, C++, Java
ML / Deep Learning	TensorFlow/Keras, PyTorch, scikit-learn, Hugging Face Transformers, Pandas, NumPy
LLM / GenAI	LangChain, OpenAI API, RAG pipelines, Prompt Engineering, Dialogflow
Backend / Serving	FastAPI, TF Serving, MySQL, PostgreSQL, React.js, React Native, MongoDB, Streamlit
Infrastructure / MLOps	GCP, Docker, GCP, CI/CD, Linux, Git
Foundation	Algorithms, Optimization, Statistics, Signal Processing

EDUCATION

Master's-level Engineering Degree in Computer Science — Sep 2019 – Sep 2024

CY Tech, CY Cergy Paris Université · France

Specialization in Artificial Intelligence · French Grande École (Master of Science in Computer Science Engineering)

Master's Thesis: Automatic Music Transcription - Designed an end-to-end system combining transformer architectures and reservoir computing to transcribe audio signals into musical scores

Baccalauréat S (A-levels equivalent) — July 2019

Lycée L'Immaculée Conception · France

ADDITIONAL INFORMATION

Languages: French (native), English (C1 · TOEIC 960), Japanese (B1), Spanish (B1), German (A1), Chinese(A1)

Leadership: President of 3 student organizations during engineering studies (Hiking, Japanese Culture, Arts) managed events and member communities of 10–20 people

Teaching: Volunteer Japanese language instructor for engineering students · 2 years

Interests: Trail, running, swimming, drawing, piano